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McLean**

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(54) **APPARATUS, SYSTEM AND/OR METHOD
FOR MODULAR FILTER SCREENS**

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E21B 21/06 (2006.01)

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CPC **B01D 29/05** (2013.01); **B01D 29/012**
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29/56 (2013.01); **B07B 1/4645** (2013.01);
E21B 21/065 (2013.01)

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B01D 29/56; B07B 1/4645; B07B
1/4627; E21B 21/065

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See application file for complete search history.

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U.S.C. 154(b) by 0 days.

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This patent is subject to a terminal dis-
claimer.

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(22) Filed: **Jul. 28, 2023**

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(65) **Prior Publication Data**

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Related U.S. Application Data

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(63) Continuation of application No. 16/737,755, filed on
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continuation of application No. 14/892,964, filed as
application No. PCT/US2014/047815 on Jul. 23,
2014, now Pat. No. 10,549,219.

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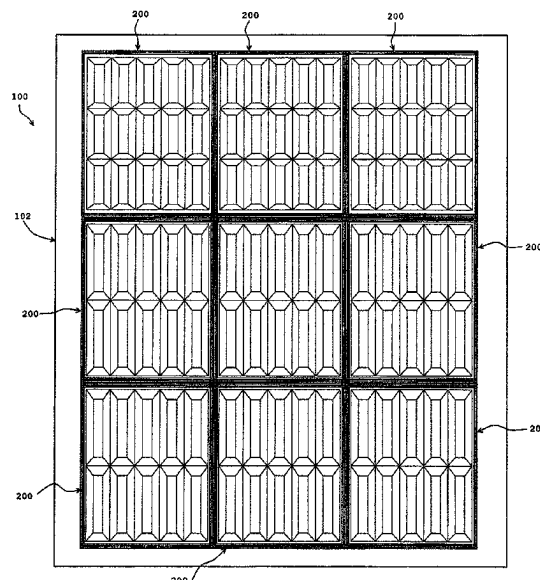
(60) Provisional application No. 61/858,538, filed on Jul.
25, 2013.

(57) **ABSTRACT**

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A method and/or a system filters particles using a screen
with modular segments installed into a frame. The frame has
a perimeter structure and an orthogonal array of support
members. The segments have a segment perimeter frame, an
orthogonal array of ribs and a mesh stretched across the
segment. When the mesh of the segments is damaged, the
damaged segment is removed from the frame and a replace-
ment segment is installed.

17 Claims, 6 Drawing Sheets



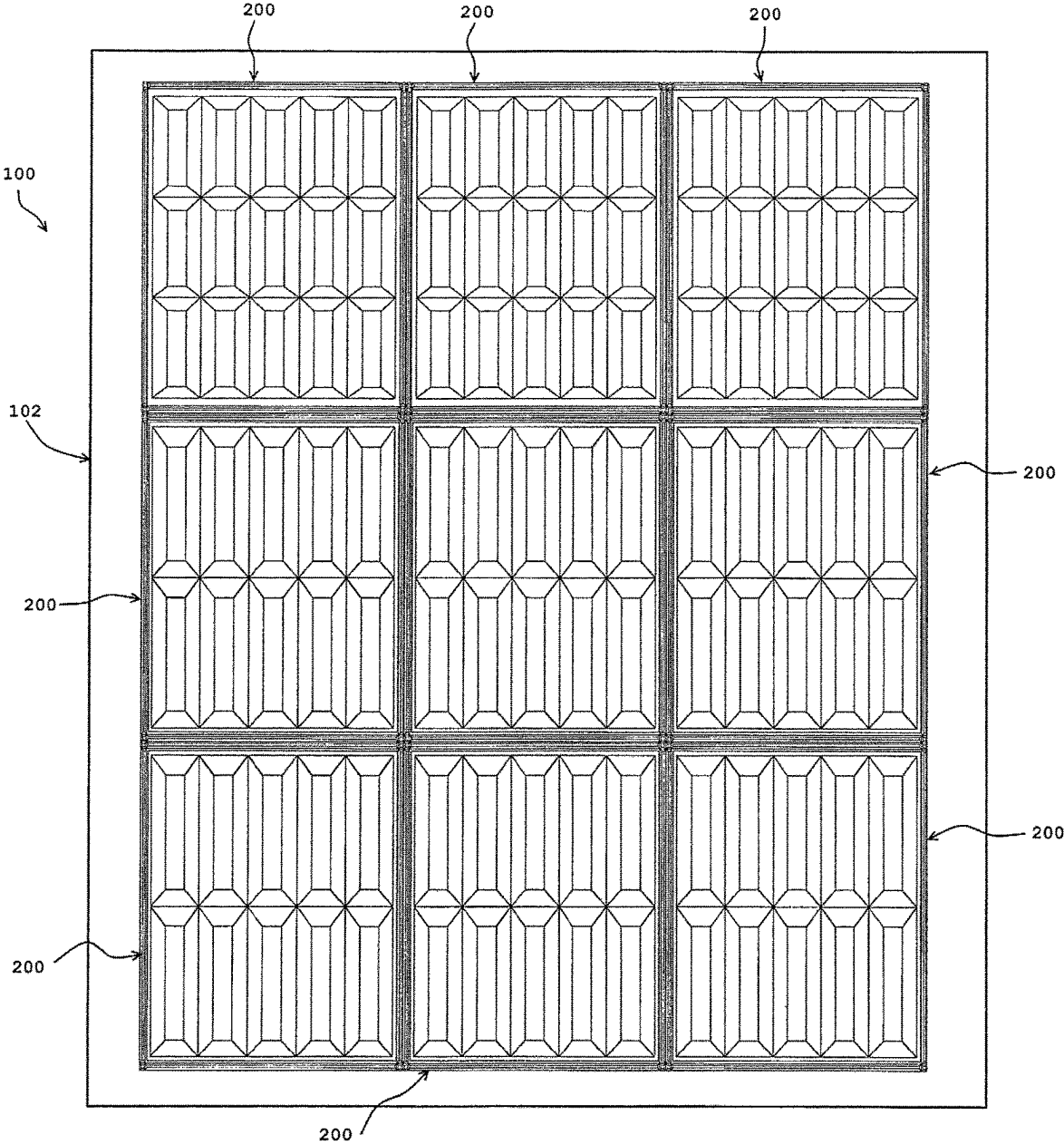


FIG. 1

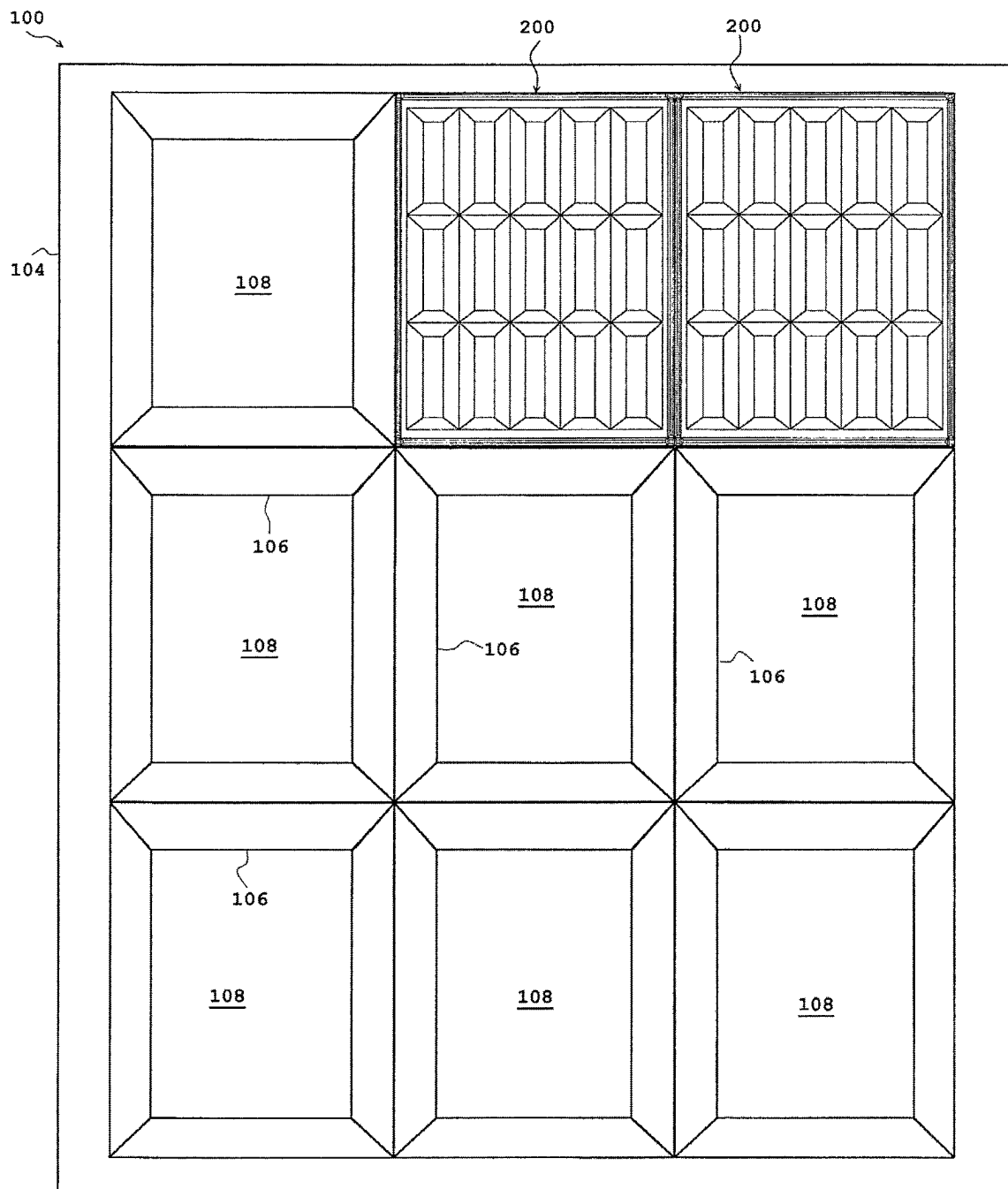


FIG. 2

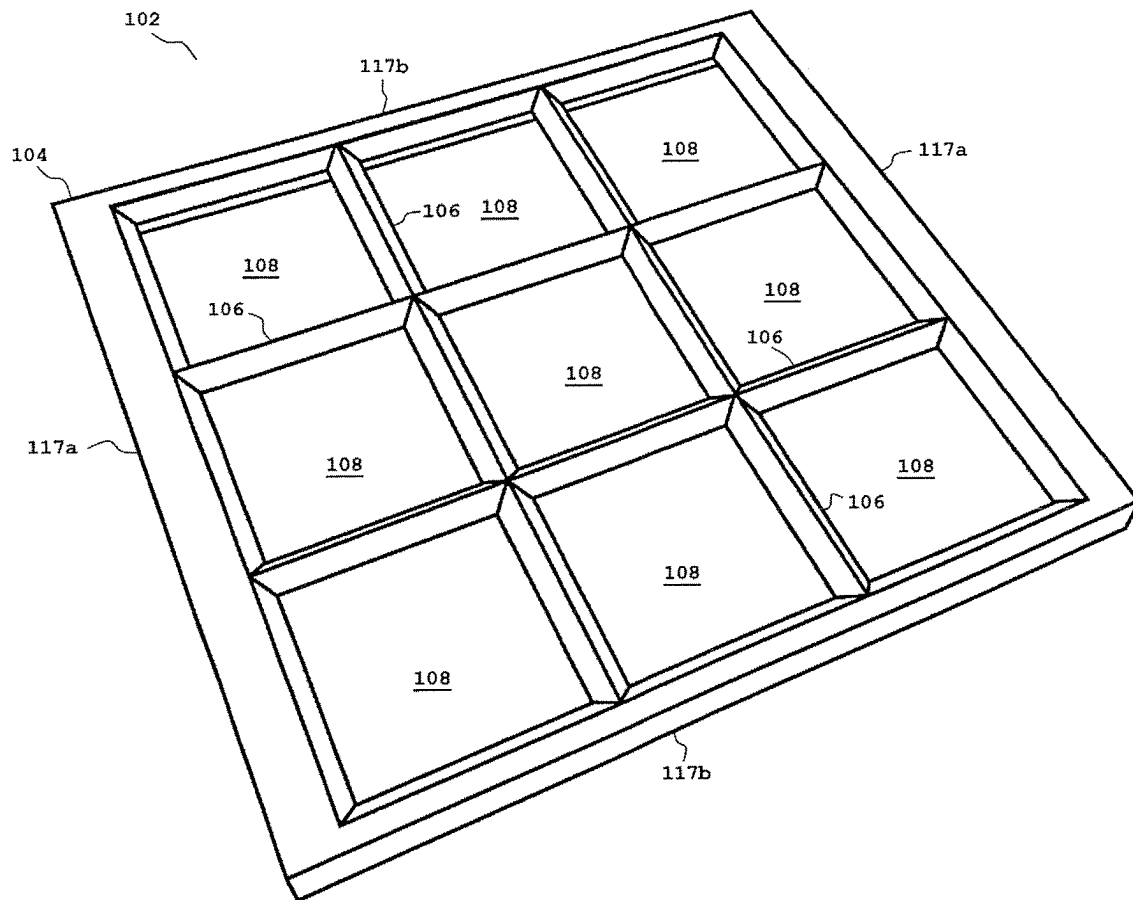


FIG. 3

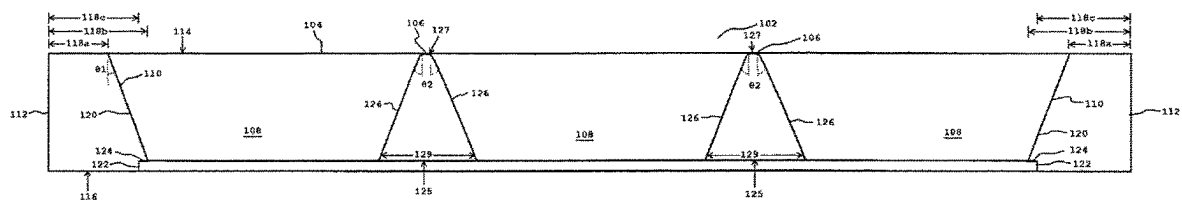


FIG. 4

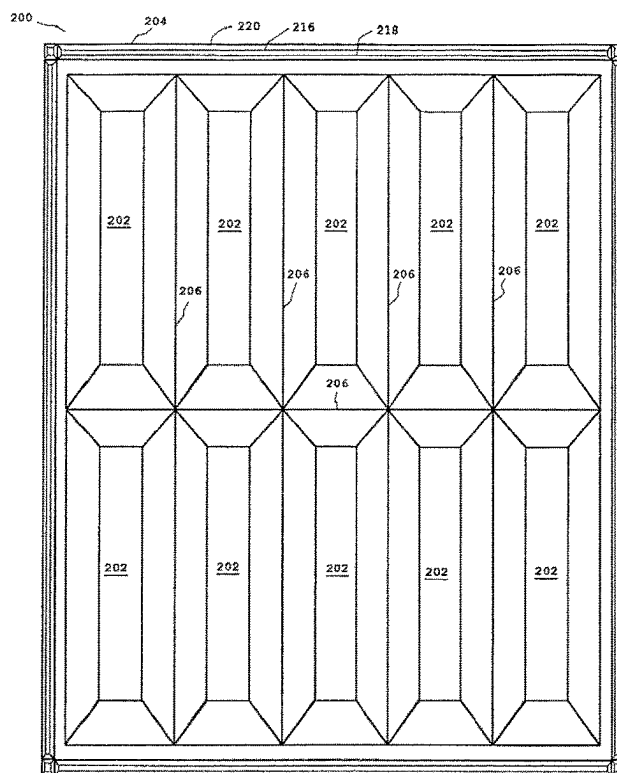


FIG. 5A

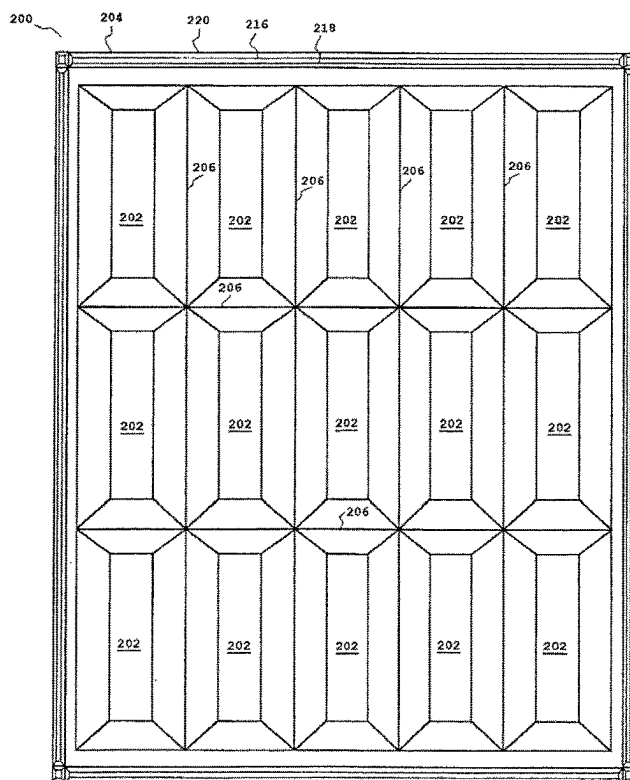


FIG. 5B

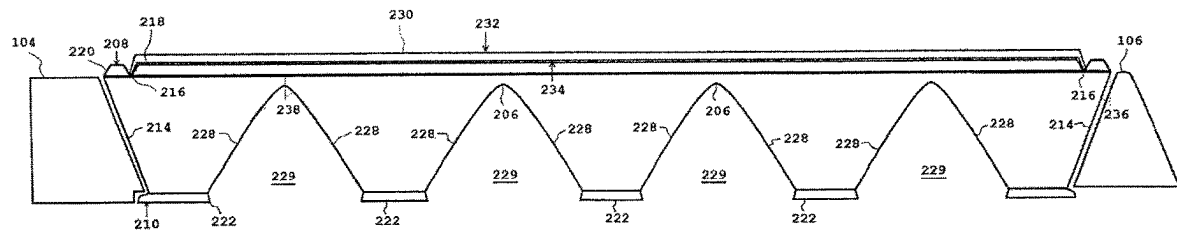


FIG. 6

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APPARATUS, SYSTEM AND/OR METHOD FOR MODULAR FILTER SCREENS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/737,755, filed on Jan. 8, 2020, which is a continuation of U.S. patent application Ser. No. 14/892,964, filed on Nov. 20, 2015, which is the National Stage of International Patent Application No. PCT/US2014/047815, filed on Jul. 23, 2014, which claims priority to U.S. Provisional Patent Application No. 61/858,538, filed on Jul. 25, 2013, the disclosures of which are incorporated herein by reference.

BACKGROUND

The embodiments disclosed herein related to an apparatus, a system and/or a method for filtering and/or separating particles. More specifically, embodiments disclosed herein relate to modular filter screens and methods using modular filter screens.

Filter screens are used to filter particles in industrial filtration systems. For example, shale shakers use filter screens to separate drill cuttings from drilling fluid in on-shore and off-shore oilfield drilling. The filters screens have a wire mesh stretched across a frame. The frame is a rectangular grid of support ribs that divide the frame into an orthogonal array of cells. The mesh is secured to the ribs as well as to the surrounding frame. Additionally, the cells in the frame have the same orientation.

The wire mesh has different mesh sizes defined by the size of the apertures between the individual wires in the mesh. The size of the apertures of the mesh is selected depending on the size of the particle to be filtered. Particles smaller than the aperture pass through the wire mesh and through the cells between the ribs. The remaining particles are discharged on an end of the filter screen.

Over the life of the filter screen, particle movement through the wire mesh, as described here, may cause wear on the wire mesh. Eventually, holes form in the wire mesh that allow larger particles to pass through the screen. Certain regions of the mesh are more prone to damage than other regions. For example, the mesh stretched across cells below where the particles are introduced experience greater wear than the mesh stretched across cells towards the exit end of the screen.

Conventional filter screens may be heavy and cumbersome to transport. Additionally, the filter screens are bulky. Locations with limited space, such as drilling rigs, may not be able to store extra filter screens. Therefore, replacement filter screens are not always available.

To extend the operational life of the screen, plugs are used to block particles from going through damaged wire mesh. These plugs fit into the cells and have a solid surface oriented towards the damaged mesh. The plugs fit into the cell from below and are hammered into place. As such, the plugs are designed to block particles from traveling through the damaged screen and the cell. Therefore, the cells with the plugs no longer filter particles.

As more of the plugs are installed into more of the cells, the filtering ability of the filter screen is reduced. Eventually, the filter screen must be replaced. Moreover, the plugs cannot be reused as they are hammered in place without a way to remove them. When the filter screen is discarded, the installed plugs are also discarded.

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Further, incorporating a design into the filter screen that allows plugs to snap into the cells increases development time and cost. Additionally, retrofitting a system so that plugs may be used to patch a hole in the mesh requires time and significant retooling.

DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a top view of an embodiment of a filter screen.

FIG. 2 illustrates a top view of an embodiment of a filter screen partially populated with segments.

FIG. 3 illustrates an isometric view of an embodiment of a frame for the filter screen.

FIG. 4 illustrates a side cutout view of an embodiment of the frame.

FIG. 5A illustrates a top view of an embodiment of one segment.

FIG. 5B illustrates a top view of an additional embodiment of one segment.

FIG. 6 illustrates a side cutout view of an embodiment of one segment.

DETAILED DESCRIPTION

The embodiments disclosed herein related to an apparatus, a system and/or a method for filtering and/or separating particles. More specifically, embodiments disclosed herein relate to modular filter screens and methods using modular filter screens.

FIG. 1 illustrates an embodiment of a screen **100**. The screen **100** may have a frame **102** and a plurality of segments **200**. FIG. 2 illustrates the screen **100** with the frame **102**. The frame **102** is only partially populated with the segments **200**. The screen **100** may have an input side **128** and an output side **130**. The input side **128** of the screen **100** may receive a material to be separated. A portion of the material may filter through the screen **100** and may be deposited in a sump, a reservoir, a hopper and/or a collection pan. A portion of the material not filtered through the screen **100** may be discharged from the output side **130** of the screen **100** and may be deposited into a shaker pit and/or other container. In an embodiment, the material to be separated may be a slurry of drilling fluid and drill cuttings. However, the screen **100** is not limited to separating solids from a liquid. The screen **100** may also separate solids of different sizes.

FIGS. 3 and 4 illustrate an embodiment of the frame **102**. As shown, the frame **102** may have a perimeter structure **104** and an orthogonal array of support members **106**. The perimeter structure **104** and the array of support members **106** may divide the frame **102** into a plurality of regions **108**. FIG. 3 illustrates an embodiment of the frame **102** with nine regions **108**. The frame **102** may have any number of regions **108** by changing the size of the perimeter structure **104** and/or changing the number of support members **106**.

The perimeter structure **104** may have an inner wall **110**, an outer wall **112**, a top **114**, a bottom **116** and a thickness **118**. The inner wall may have a tapered portion **120**, a flat portion **122** and a lip **124**. The tapered portion **120** of the inner wall **110** may taper downwardly toward the center of the perimeter structure **104** at an angle **132**. The top **114** of the perimeter structure **104** may have a top thickness **118a**. As the tapered portion **120** of the inner wall **110** tapers downwardly, the thickness **118** of the perimeter structure **104** may increase until the thickness **118** equals an intermediate thickness **118b**. The flat portion of the inner wall may

have a uniform bottom thickness **118c**. The intermediate thickness **118b** may be greater than the top thickness **118a**. The intermediate thickness **118b** may be greater than the bottom thickness **118c**. The lip **124** may form at an interface **123** between the tapered portion **120** and the flat portion **122** of the inner wall **110**.

The outer wall **112** of the perimeter structure **104** may be flat. The perimeter structure **104** may be rectangular with secure sides **117a** and support sides **117b**. The top thickness **118a**, the intermediate thickness **118b** and the bottom thickness **118c** of the secure sides **117a** may be greater than the top thickness **118a**, the intermediate thickness **118b** and the bottom thickness **118c** respectively of the support sides **117b**. In an embodiment, the secure sides **117a** may be longer than the support sides **117b**. The perimeter structure **104** may be configured to fit in a shaker (not shown) and may be secured into place along the secure sides **117a**. In an embodiment, the top thickness **118a**, the intermediate thickness **118b** and the bottom thickness **118c** of the secure sides **117a** may be equal to the top thickness **118a**, the intermediate thickness **118b** and the bottom thickness **118c** respectively of the support sides **117b**. In an embodiment, the perimeter structure **104** may be secured along the secure sides **117a** and the support sides **117b**. The perimeter structure **104** may be secured into a filtration system (not shown) by a wedge, clamp or any other means known to a person of ordinary skill in the art. In an embodiment, the screen **100** may be secured in a basket (not shown) of the filtration system via the perimeter structure **104**.

The support members **106** may have a bottom **125**, sides **126** and a top **127**. The sides **126** of each of the support members **106** may join at the top **127** of the support member **106** and may taper outwardly and downwardly at the angle **134**. The angle **134** of the sides **126** of the support members **106** may equal the angle **132** of the tapered portion **120** of the inner wall **110** of the perimeter structure **104**. Each of the support members **106** may have a thickness **129** at the bottom of each of the support members **106**. The support members **106** may interface with the tapered portion **120** of the inner wall **110** of the perimeter structure **104** to form the plurality of regions **108**. The lip **124** of the inner wall **110** of the perimeter structure **104** may be coplanar with the bottom **125** of the support members **106**.

The regions **108** may have rectangular cross-sections. As the tapered portion **120** of the inner wall **110** of the perimeter structure **104** and the sides **126** of the support members **106** taper downwardly, the cross-sections of the plurality of regions **108** may decrease in size.

FIG. 5A illustrates an embodiment of one of the plurality of segments **200**. Each of the segments **200** may be divided into a plurality of cells **202** by a segment perimeter frame **204** and an orthogonal array of ribs **206**. FIG. 5A shows the segments **200** with ten cells **202**. However, the segment **200** may have more cells **202** or less cells **202** as required by an embodiment of the segment **200**. The segments **200** may have any number of the cells **202** by changing the number of the ribs **206**. For example, in an embodiment illustrated in FIG. 5B, the segments **200** may have fifteen cells **202**.

The segment perimeter frame **204** may have an outer wall **214**, a top **208** and a bottom **210**. The top **208** of the segment perimeter frame **204** may have a groove **216** defining an inner perimeter **218** and an outer perimeter **220**. As shown in FIG. 6, the segment perimeter frame **204** may have a flange **222** extending outwardly from the outer wall **212** at the bottom **210** of the segment **200**. The outer wall **212** may taper inwardly from the top **208** of the segment **200** to the bottom **210** of the segment **200** at an angle **236**. The angle

236 of the outer wall **214** may equal the angle **132** of the tapered portion **120** of the inner wall **110** of the perimeter structure **104** and the sides **126** of the support members **106**.

Each of the cells **202** may have walls **228** defined by the segment perimeter frame **204** and the orthogonal array of ribs **206**. The walls **228** of the cells **202** may taper downwardly and inwardly at the angle **238** towards the center of each of the cells **202**. Each of the cells **202** may have a rectangular cross-section. As the walls **228** of each of the cells **202** taper downwardly and inwardly, the cross-section of each of the cells **202** may be smaller at the bottom **210** of the cell **202** compared to the top **208** of the cell **202**.

Each of the ribs **206** may define a cross-sectional area. In an embodiment, the cross-sectional area of each of the ribs **206** may be a triangle. Additionally, each of the ribs **206** may be hollow. Alternatively, in an embodiment, the cross-section of each of the ribs **206** may be an inverted V with a space **229** between the walls **228** of the cells **202**. The outer wall **212** of the segment perimeter frame **204** may be configured to expose the space **229** between the walls **228** of the cells **202**.

As shown in FIG. 6, a mesh **230** may be stretched across the inner perimeter **218** of the segment perimeter frame **204** and may be secured to the groove **216**. The mesh **230** may have a top-side **232** and an underside **234**. The underside **234** of the mesh **230** may be supported by the inner perimeter **218** of the segment perimeter frame **204** and the top **208** of the orthogonal array of ribs **206**. The mesh **230** may be a single layer of woven mesh wire or multiple layers of woven mesh wire. The mesh **230** may have a mesh size to filter particles for an embodiment. For example, the mesh **230** may have the mesh size to separate drill cuttings from circulated drill fluid. The mesh size as used herein refers to the size of the apertures in the mesh **230**.

The segments **200** may have different configurations. Between the different configurations, the segments **200** may have cells **202** with different sizes and/or may have mesh **230** with different mesh sizes. For example, the segments **200** that may be installed into the frame **102** in areas of heavy wear, such as the segments **200** closer to the input **128** of the screen **100**, may have smaller sized cells **202** compared to segments **200** closer to the output **130** of the screen **100**. The screens **100** with smaller cells **202a** may be more durable compared to screens **100** with larger cells **202b**. The segment perimeter frame **204** and/or the orthogonal array of ribs **206** may have a color. Different configurations of the segments **200** may have the segment perimeter frame **204** and/or the orthogonal array of ribs **206** of different colors.

The segments **200** may be installed into the frame **102**. One of the segments **200** may be inserted into one of the regions **108** so that the segment **200** fits within the geometry of the region **108**. The flange **222** of the segment **200** that may be installed may flex during insertion into one of the regions **108** of the frame **102**. The flange **222** may interface with the lip **124** of the perimeter structure **104** and the bottom **125** of the array of support members **106** to lock the segment **200** installed into the region **108**.

FIG. 1 shows an embodiment of the screen **100** with the segments **200** installed into the regions **108** of the frame **102**. Additionally, FIG. 1 shows a screen **100** with the segments **200** having different configurations. In FIG. 1, the segments **200** adjacent to the input side **128** of the screen **100** may be configured as the segments **200** shown in FIG. 5A. The remaining segments **200** are configured as the segments **200** shown in FIG. 5B. However, the configuration of the segments **200** installed into the frame **102** may be different than

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the configuration shown in FIG. 1. The segments 200 may be installed by hand or with a tool, such as a hammer or a rubber mallet.

Each of the segments 200 may have an orientation with respect to the alignment of the cells 202. FIG. 1 shows an embodiment where the segments 200 have the same orientation. However, some of the segments 200 may have different orientations as installed. In an embodiment, some of the segments 200 may be rotated 90 degrees with respect to the rest of the segments 200.

The frame 102 may be designed to be durable so that the frame 102 may be used during the operational life of a shale shaker. Conversely, the segments 200 may be designed to be disposable. When the mesh 230 on one of the segments 200 is damaged, the segment 200 may be replaced and may be discarded, recycled and/or refurbished.

When designing the screen 100 for a particular application, the segments 200 installed into the frame 102 may be chosen from segments 200 with different configurations. The screen 100 may be shipped to a location with the segments 200 separate from the frame 102. To facilitate repair, the replacement segments 200 may also be shipped to the location. The segments 200 may be installed into the frame 102 and the screen 100 may be installed into the filtration system.

After the screen 100 is installed, the filtration system may be operated. The segments 200 adjacent to the input side 128 of the screen 100 may become damaged more frequently than the other segments 200 in the screen 100. When the mesh 230 of one or more of the segments 200 is damaged, the operation of the filtration system may be halted. The screen 100 may be removed from the filtration system. The segment 200 with the damaged mesh 230 may be removed by pushing the bottom 210 of the segment 200 until the flange 222 of the segment 200 disengages from the lip 124 of the inner wall 110 of the perimeter structure 104 and the bottom 125 of the orthogonal array of support members 106. The removed segment 200 may be discarded, recycled and/or refurbished. The removed segment 200 may be replaced and the screen 100 may then be reinstalled into the filtration system.

While the present disclosure has been described with respect to a limited number of embodiments, those skilled in the art, having benefit of this disclosure, will appreciate that other embodiments may be devised which do not depart from the scope of the disclosure as described herein. Accordingly, the scope of the present disclosure should be limited only by the attached claims.

What is claimed is:

1. A screen, comprising:

a frame, including:

a perimeter structure comprising a top perimeter surface, a bottom perimeter surface, and an inner perimeter wall, wherein the bottom perimeter surface is a bottom extremity of the perimeter structure;

a lip formed in the inner perimeter wall between the inner perimeter wall and the bottom perimeter surface; and

a support member connected to the perimeter structure, wherein the support member comprises a top support surface, a bottom support surface, a first support side, and a second support side, and wherein the first support side and the second support side oppose one another and are flat from the top support surface to the bottom support surface, wherein the bottom perimeter surface extends lower than the bottom support surface, and wherein the inner perimeter

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wall and one or more of the first support side or the second support side form an opening; and

a segment positioned in the opening and removably connected to the frame, the segment including:

a segment perimeter frame having a top segment surface, a bottom segment surface, an outer segment wall, and an inner segment wall, wherein the outer segment wall contacts the inner perimeter wall and the inner segment wall contacts the first support side or the second support side;

a first flange extending from the outer segment wall at the bottom segment surface and a second flange extending from the inner segment wall at the bottom segment surface, the bottom segment surface being a lowest extremity of the segment perimeter frame, wherein the first flange extends at least partially into the lip formed in the inner perimeter wall and the second flange extends beneath the bottom support surface, to removably connect the segment to the frame; and

a rib connected to the segment perimeter frame, wherein the segment perimeter frame and the rib define a plurality of cells of the segment;

wherein the first flange and the second flange define the bottom segment surface, wherein the bottom segment surface extends below the bottom support surface, and wherein the bottom segment surface does not extend below the bottom perimeter surface.

2. The screen of claim 1, wherein the first support side and the second support side taper such that a thickness of the support member increases proceeding from the top support surface to the bottom support surface.

3. The screen of claim 1, wherein the support member has a substantially triangular cross-sectional shape.

4. The screen of claim 1, wherein the inner perimeter wall comprises a tapered portion, and wherein a thickness of the perimeter structure increases proceeding downward along the tapered portion from a first thickness to a second thickness.

5. The screen of claim 4, wherein the lip further comprises a lip portion and a flat portion between the inner perimeter wall and the bottom perimeter surface.

6. The screen of claim 5, wherein the thickness of the perimeter structure decreases proceeding along the lip portion toward the flat portion from the second thickness to a third thickness.

7. The screen of claim 6, wherein the third thickness is greater than the first thickness and less than the second thickness.

8. The screen of claim 5, wherein the lip portion defines a lip surface that is substantially coplanar with the bottom support surface in a horizontal plane.

9. The screen of claim 1, wherein the perimeter structure comprises a secure side and a support side, wherein a thickness of the secure side is greater than a thickness of the support side.

10. The screen of claim 9, wherein the secure side is longer than the support side, and wherein the perimeter structure is configured to be positioned at least partially within a shaker and secured within the shaker by the secure side but not by the support side.

11. A method comprising:

arranging a plurality of screen modules having a first plurality of cells within regions of the frame of claim 1 wherein the regions are defined by a perimeter wherein the plurality of screen modules have cells covered by a first mesh;

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removing a first screen module of the plurality of screen modules from the frame of claim 1 to create a vacant region in the frame of claim 1; and

inserting a second screen module of the plurality of screen modules into the vacant region of the frame of claim 1.

12. The method of claim 11 further comprising:

inserting a third screen module of the plurality of screen modules into the frame of claim 1 wherein the third screen module is arranged in a direction perpendicular to the frame of claim 1.

13. The method of claim 11 further comprising:

inserting a fourth screen module of the plurality of screen modules into the frame of claim 1 wherein the fourth screen module has a second plurality of cells wherein the cells of the second plurality of cells are different sizes than the cells of the first plurality of cells.

14. The method of claim 11 further comprising:

inserting a replacement screen module into the frame of claim 1 wherein the replacement screen module is

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covered by a second mesh with a second mesh density wherein the second mesh density is different than a density of the first mesh.

15. A method comprising:

placing support members on the frame of claim 1 with the perimeter structure of claim 1 to define a region;

inserting a screen segment with a flange into the region; and

interlocking the flange of the screen segment with the perimeter structure of claim 1.

16. The method of claim 15 further comprising:

flexing the flange of the screen segment during insertion into the region of the frame of claim 1.

17. The method of claim 15 further comprising:

securing the screen segment into the region of the frame of claim 1 via an interference fit; or

interlocking the flange of the screen segment with the support members of claim 1.

* * * * *